

Ex Demonstrați că la euclidean am origine și de rază 1, corp. normarea $\|\cdot\|_1$ și $\|\cdot\|_\infty$ din $\mathbb{R}^2$.

Sol:

$\forall x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, avem:

- $\|x\|_1 = |x_1| + |x_2| + \dots + |x_n|$

- $\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$

- $\|x\|_\infty = \max\{|x_1|, |x_2|, \dots, |x_n|\}$

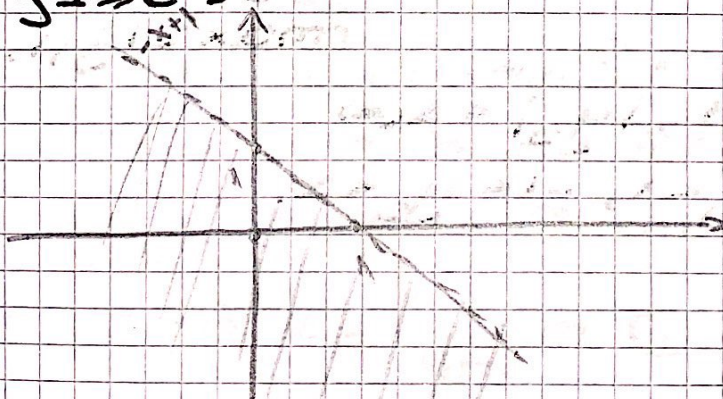
Pentru $\alpha > 0$ $B_\alpha(x, \alpha) = \{y \in \mathbb{R}^2 \mid \|y - x\|_\infty < \alpha\}$
 la noi: $\mathbb{R}^2$, $x = (0, 0)$, $\alpha = 1$, $\|\cdot\|_1$, $\|\cdot\|_\infty$

$B_1((0, 0), 1) = \{y \in \mathbb{R}^2 \mid \|(y_1, y_2) - (0, 0)\|_\infty < 1\}$

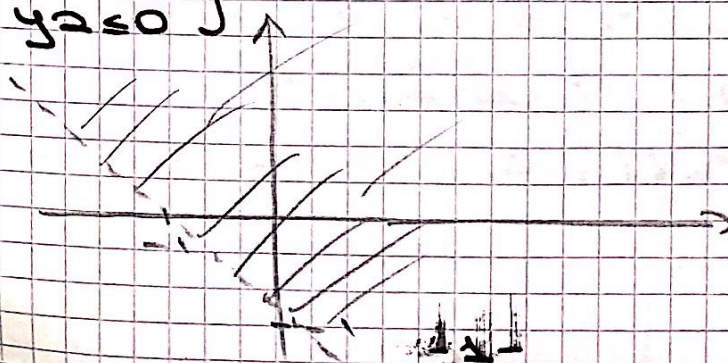
$\|(y_1, y_2)\|_\infty < 1$

$|y_1| + |y_2| < 1$

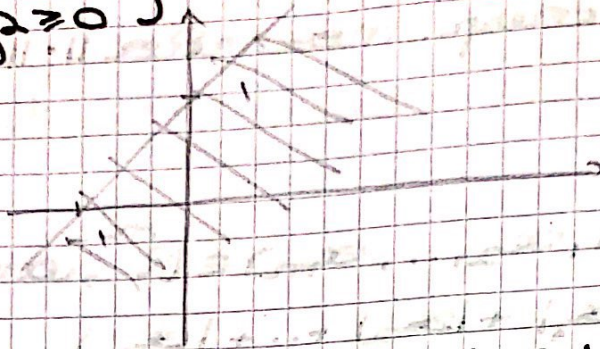
- $\left. \begin{matrix} y_1 \geq 0 \\ y_2 \geq 0 \end{matrix} \right\} \Rightarrow y_1 + y_2 < 1 \Rightarrow y_2 < 1 - y_1$



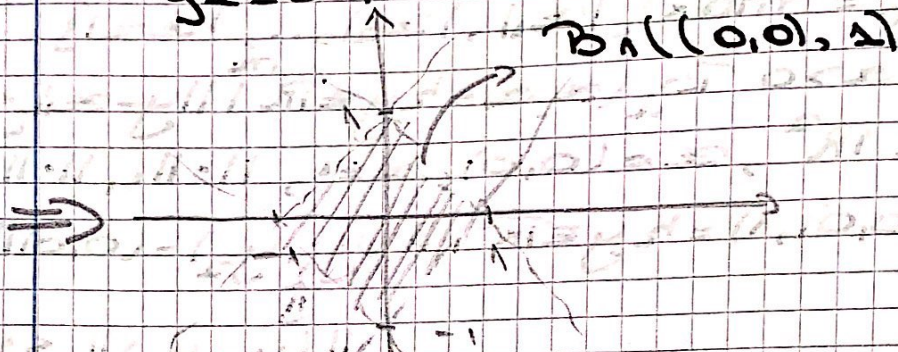
- $\left. \begin{matrix} y_1 \leq 0 \\ y_2 \leq 0 \end{matrix} \right\} \Rightarrow -y_1 - y_2 < 1 \Rightarrow y_2 > -y_1 - 1$



$$\begin{aligned} \bullet \quad & \begin{cases} y_1 \leq 0 \\ y_2 \geq 0 \end{cases} \Rightarrow -y_1 + y_2 < 1 \Rightarrow y_2 < 1 + y_1 \end{aligned}$$



$$\bullet \quad \begin{cases} y_1 \geq 0 \\ y_2 \leq 0 \end{cases} \Rightarrow y_1 - y_2 < 1 \Rightarrow y_2 > y_1 - 1$$



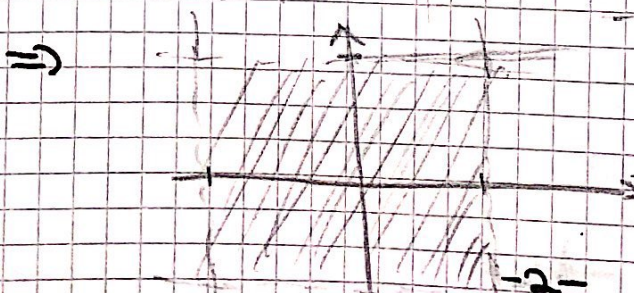
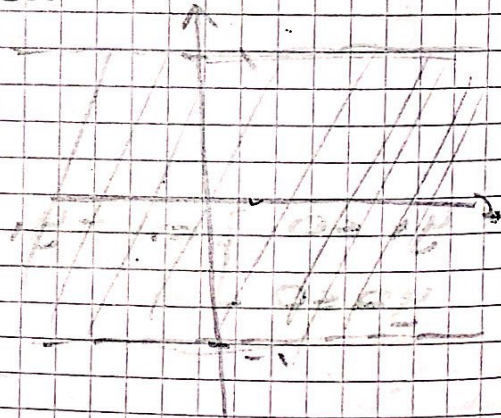
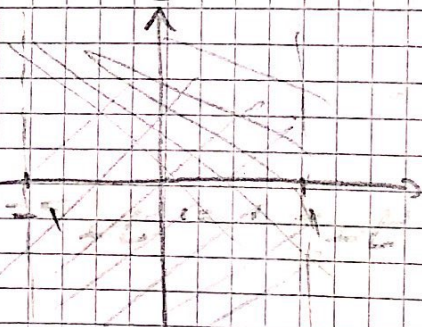
$$\bullet \quad B_\infty((0,0),1) = \{y \in \mathbb{R}^2 \mid \|(y_1, y_2) - (0,0)\|_\infty < 1\}$$

$$\Leftrightarrow \|(y_1, y_2)\|_\infty < 1$$

$$\max\{|y_1|, |y_2|\} < 1$$

$$\Leftrightarrow |y_1| < 1, |y_2| < 1 \Leftrightarrow$$

$$\Leftrightarrow -1 < y_1 < 1, -1 < y_2 < 1.$$



Ex:

$$\left. \begin{aligned} \text{a) } \mathcal{B} \in \mathbb{R}^n : \|\cdot\|_1 \sim \|\cdot\|_2 \\ \text{b) } \mathcal{B} \in \mathbb{R}^n : \|\cdot\|_1 \sim \|\cdot\|_\infty \\ \text{c) } \mathcal{B} \in \mathbb{R}^n : \|\cdot\|_2 \sim \|\cdot\|_\infty \end{aligned} \right\} \Rightarrow \|\cdot\|_1 \sim \|\cdot\|_2 \sim \|\cdot\|_\infty$$

$$\|\cdot\|_i \sim \|\cdot\|_j \Leftrightarrow \exists m, n > 0 \text{ s.t.}$$

$$m \cdot \|\cdot\|_i \leq \|\cdot\|_j \leq n \cdot \|\cdot\|_i$$

SOL:

$$\text{a) } \|\cdot\|_1 \sim \|\cdot\|_2 \Leftrightarrow \exists \alpha, \beta > 0 \text{ s.t.}$$

$$\alpha \cdot \|\vec{x}\|_1 \leq \|\vec{x}\|_2 \leq \beta \|\vec{x}\|_1, \forall \vec{x} \in \mathbb{R}^n$$

$$\alpha (|\vec{x}_1| + |\vec{x}_2| + \dots + |\vec{x}_n|) \leq \sqrt{\vec{x}_1^2 + \dots + \vec{x}_n^2}$$

$$\stackrel{\textcircled{1}}{\Rightarrow} \beta (|\vec{x}_1| + |\vec{x}_2| + \dots + |\vec{x}_n|), \forall \vec{x}_i \in \mathbb{R}$$

C.B.S $\therefore |\langle \vec{x}, \vec{y} \rangle| \leq \|\vec{x}\|_2 \cdot \|\vec{y}\|_2, \forall \vec{x}, \vec{y} \in \mathbb{R}^n$

$$(\vec{x}_1 \vec{y}_1 + \vec{x}_2 \vec{y}_2 + \dots + \vec{x}_n \vec{y}_n)^2 =$$

$$= (\vec{x}_1^2 + \vec{x}_2^2 + \dots + \vec{x}_n^2)(\vec{y}_1^2 + \dots + \vec{y}_n^2)$$

• $\beta \neq \beta = 1 \Rightarrow$ evid $\textcircled{2}$ undero.

• $\beta \neq \textcircled{1}$: In C.B.S. wärgem:

$$\left\{ \begin{aligned} \vec{x}_k &= 1, \forall k = \overline{1, n} \\ \vec{y}_k &= |\vec{x}_k| \end{aligned} \right.$$

$$\Rightarrow (1 \cdot |\vec{x}_1| + \dots + 1 \cdot |\vec{x}_n|)^2 \leq n \cdot (\vec{x}_1^2 + \dots + \vec{x}_n^2)$$

$$\Rightarrow |\vec{x}_1| + \dots + |\vec{x}_n| \leq \sqrt{n} \cdot \sqrt{\vec{x}_1^2 + \dots + \vec{x}_n^2}$$

$$\Rightarrow \frac{1}{\sqrt{n}} (|\vec{x}_1| + \dots + |\vec{x}_n|) \leq \sqrt{\vec{x}_1^2 + \dots + \vec{x}_n^2}$$

offgemärd $\alpha = \frac{1}{\sqrt{n}} \Rightarrow \textcircled{1}$ underoadd

$$\text{b) } \|\cdot\|_1 \sim \|\cdot\|_\infty \Leftrightarrow \exists a, b > 0 \text{ s.t.}$$

$$a \cdot \|\cdot\|_1 \leq \|\cdot\|_\infty \leq b \cdot \|\cdot\|_1, \Leftrightarrow$$

$$\Rightarrow a (|\vec{x}_1| + \dots + |\vec{x}_n|) \leq \max_{i=1, \dots, n} |\vec{x}_i| + \dots + |\vec{x}_n|$$

$$= b (|\vec{x}_1| + \dots + |\vec{x}_n|), \forall \vec{x}_i \in \mathbb{R}$$

- Offensichtl. $\|x\|_1 \sim \|x\|_\infty$
- Zz. ①: Asem $\forall i=1, \dots, m$,
 $|x_i| \leq \max\{|x_1|, \dots, |x_m|\} \Rightarrow$
 $\Rightarrow |x_1| + \dots + |x_m| \leq m \cdot \max\{|x_1|, \dots, |x_m|\}$
 Offensichtl. $\alpha = \frac{1}{m} \Rightarrow$ ① adeq.

2) $\|\cdot\|_2 \sim \|\cdot\|_\infty \Leftrightarrow$
 $\Rightarrow \exists A, B > 0$ mit $A \cdot \|\cdot\|_\infty \leq \|\cdot\|_2 \leq B \cdot \|\cdot\|_\infty$
 Dim 1) $\Rightarrow \exists \alpha, \beta > 0$ mit $\alpha \cdot \|\cdot\|_1 \leq \|\cdot\|_2 \leq \beta \cdot \|\cdot\|_1$
 Dim 2) $\Rightarrow \exists \alpha, \beta > 0$ mit $\alpha \cdot \|\cdot\|_1 \leq \|\cdot\|_\infty \leq \beta \cdot \|\cdot\|_1$

$\Rightarrow \left\{ \begin{aligned} \|\cdot\|_2 &\geq \alpha \cdot \|\cdot\|_1 \geq \alpha \cdot \frac{1}{\beta} \cdot \|\cdot\|_\infty = \\ &= \frac{\alpha}{\beta} \cdot \|\cdot\|_\infty = A \cdot \|\cdot\|_\infty \\ \|\cdot\|_2 &\leq \beta \cdot \|\cdot\|_1 \leq \beta \cdot \frac{1}{\alpha} \cdot \|\cdot\|_\infty = \\ &= \frac{\beta}{\alpha} \cdot \|\cdot\|_\infty = B \cdot \|\cdot\|_\infty \end{aligned} \right\}$

$\Rightarrow A \cdot \|\cdot\|_\infty \leq \|\cdot\|_2 \leq B \cdot \|\cdot\|_\infty, \forall x \in \mathbb{R}^n$

3. Für x so auch für $\|\cdot\|_1$ und $\|\cdot\|_2$ dass
 normen $\sim$ für X . Dem. via de generau-
 ză necesari de topologie pe X .

Sol.

Dim ap. $\|\cdot\|_1 \sim \|\cdot\|_2 \stackrel{\text{def}}{\Rightarrow}$
 $\Rightarrow \exists \alpha, \beta > 0$ mit $\alpha \cdot \|\cdot\|_1 \leq \|\cdot\|_2 \leq \beta \cdot \|\cdot\|_1$
 $\forall x \in X$ (*)

Für τ_1, τ_2 topologie pe X .

Fix $a \in X, \varepsilon > 0$, Asem:

$B_1(a, \varepsilon) := \{x \in X \mid \|x - a\|_1 < \varepsilon\}$

$B_2(a, \varepsilon) := \{x \in X \mid \|x - a\|_2 < \varepsilon\}$

Arătăm că $B_1(a, \frac{\rho}{\beta}) \subseteq B_2(a, \rho)$ (**)

• Fie $x \in B_1(a, \frac{\rho}{\beta}) \Rightarrow$
 $\Rightarrow \|x - a\|_1 < \frac{\rho}{\beta} \stackrel{(*)}{\Rightarrow} \|x - a\|_2 \leq \beta \cdot \|x - a\|_1$
 $< \beta \cdot \frac{\rho}{\beta} = \rho \Rightarrow$
 $\Rightarrow x \in B_2(a, \rho) \Rightarrow (**)$ dem.

• Fie $D \in \mathcal{T}_2$. Dem. dem. că $D \in \mathcal{T}_1$.

Dim. ip. $D \in \mathcal{T}_2 \Rightarrow$
 $\Rightarrow \forall a \in D, \exists \alpha_a > 0$ a.t. $B_2(a, \alpha_a) \subset D$
 $\stackrel{(**)}{\Rightarrow} B_1(a, \frac{\alpha_a}{\beta}) \subseteq B_2(a, \alpha_a) \subset D \Rightarrow$
 $\Rightarrow D \in \mathcal{T}_1$.

Am arătat că dacă $D \in \mathcal{T}_2 \Rightarrow$
 $\Rightarrow D \in \mathcal{T}_1$.

Prim vom avea $\mathcal{T}_1 = \mathcal{T}_2$.



Derivata după o direcție. Derivate
 parțiale. Diferențiabilityte

Def. Fie $f: D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, $a \in D$, $v \in \mathbb{R}^n$.

Un vector $L_v \in \mathbb{R}^m$ s.m. derivata
~~parțială~~ f în pct. a , după
 vectorul v , dacă:

$$\exists \text{ și limita } \lim_{t \rightarrow 0} \frac{f(a + tv) - f(a)}{t} \stackrel{\text{not}}{=} L_v$$

$$= f_v(a) = \frac{df}{dv}(a).$$

Obs. Fie $v = e_i = (0, 0, \dots, \overset{i}{1}, 0, \dots, 0) \Rightarrow$

$$\Rightarrow \frac{df}{de_i}(a) = \frac{df}{dx_i}(a) \text{ s.m. derivata}$$

parțială a lui f în raport cu
 var. x_i în pct. a .

Se mai poate calcula si astfel:

$$\frac{df}{dx_i}(a_1, \dots, a_i, \dots, a_p) = \lim_{x_i \rightarrow a_i} \frac{f(a_1, a_2, \dots, x_i, \dots, a_p) - f(a_1, a_2, \dots, a_i, \dots, a_p)}{x_i - a_i}$$

(R2)

$$\frac{df}{dx}(a, b) = \lim_{x \rightarrow a} \frac{f(x, b) - f(a, b)}{x - a}$$

$$\left. \begin{array}{l} \frac{df}{dx_1}(a, b) \\ \frac{df}{dx_2}(a, b) \\ \vdots \end{array} \right\}$$

$$\frac{df}{dy}(a, b) = \lim_{y \rightarrow b} \frac{f(a, y) - f(a, b)}{y - b}$$

$$\left. \begin{array}{l} \frac{df}{dy_1}(a, b) \\ \frac{df}{dy_2}(a, b) \\ \vdots \end{array} \right\}$$

Exp: Calculeaza toate derivatele parțiale

$$\frac{df}{dx}(-1, 1), \frac{df}{dy}(-1, 1)$$

SOL:

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$f(x, y) = 2x^2y - e^{x^2}$$

$$\frac{df}{dy}(-1, 1)$$

$$\begin{aligned} 1) \frac{df}{dy}(-1, 1) &= \lim_{t \rightarrow 0} \frac{f((-1, 1) + t \cdot (0, 1)) - f(-1, 1)}{t} \\ &= \lim_{t \rightarrow 0} \frac{f(-1, 1+t) - f(-1, 1)}{t} \\ &= \lim_{t \rightarrow 0} \frac{-2(1+t) + 2 + e - e}{t} = \\ &= -2 \end{aligned}$$

$$\begin{aligned} 2) \frac{df}{dx}(-1, 1) &= \lim_{y \rightarrow 1} \frac{f(-1, y) - f(-1, 1)}{y - 1} \\ &= \lim_{y \rightarrow 1} \frac{-2y - e + 2 + e}{y - 1} = -2 \end{aligned}$$

$$3) \frac{\partial f}{\partial x}(x, y) = 2xz^2 \Rightarrow$$

$$\Rightarrow \frac{\partial f}{\partial y}(-1, 1) = -2.$$

2. ? $\frac{\partial f}{\partial x}(0, 0) \quad \frac{\partial f}{\partial y}(0, 0).$

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \sqrt{x^2 + y^2}.$$

$$\lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} =$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{x^2}}{x} = \lim_{x \rightarrow 0} \frac{|x|}{x} < \frac{1}{1}$$

$$\Rightarrow \nexists \frac{\partial f}{\partial x}(0, 0)$$

Analog $\nexists \frac{\partial f}{\partial y}(0, 0).$

Diferențialitate:

Fie $f: D \subseteq \mathbb{R}^p \rightarrow \mathbb{R}^q$, $a \in D$ un pct.
Spunem că f este diferențiable
în a dacă:

$\exists$ o ap. liniară $L: \mathbb{R}^p \rightarrow \mathbb{R}^q$ ai.

$$\lim_{x \rightarrow a} \frac{f(x) - f(a) - L(x-a)}{\|x-a\|} = 0$$

De $\exists$, $L := Df(a) = df(a) = f'(a)$ și
sm. diferențiale) f în a .

nu este o constantă!

Forma diferențialei:

$$f: \mathbb{R}^p \rightarrow \mathbb{R} \begin{cases} \rightarrow f: \mathbb{R} \rightarrow \mathbb{R}: Df(a) = f'(a) \cdot x \\ Df(a)(u) = f'(a) \cdot u \\ \Rightarrow f: \mathbb{R}^2 \rightarrow \mathbb{R}: Df(a) = \frac{\partial f}{\partial x}(a) dx + \frac{\partial f}{\partial y}(a) dy \\ Df(a)(u) = \frac{\partial f}{\partial x}(a) \cdot u_1 + \frac{\partial f}{\partial y}(a) \cdot u_2 \end{cases}$$

$$f: \mathbb{R}^p \rightarrow \mathbb{R}$$

$$D_f(a) = \frac{\partial f}{\partial x_1}(a) dx_1 + \dots + \frac{\partial f}{\partial x_p}(a) dx_p$$

$$D_f(a)(u) = \frac{\partial f}{\partial x_1}(a) \cdot u_1 + \dots + \frac{\partial f}{\partial x_p}(a) \cdot u_p$$

$(u_1, \dots, u_p)$

$$\bullet \quad f: \mathbb{R} \rightarrow \mathbb{R}^q \quad (p=1) \quad \text{let } f(x) = (f_1(x), f_2(x), \dots, f_q(x))$$

$$D_f(a) = (D_{f_1}(a), D_{f_2}(a), \dots, D_{f_q}(a)) =$$

$$= (f'_1(a) \cdot x, f'_2(a) \cdot x, \dots, f'_q(a) \cdot x)$$

$$\Rightarrow D_f(a)(u) = (f'_1(a) \cdot u, f'_2(a) \cdot u, \dots, f'_q(a) \cdot u)$$

$$\bullet \quad f: \mathbb{R}^p \rightarrow \mathbb{R}^q$$

$(p, q > 1)$

$\Rightarrow$ def. matrix JACOBIAN:

$$J_f(a) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(a) & \dots & \frac{\partial f_1}{\partial x_p}(a) \\ \frac{\partial f_2}{\partial x_1}(a) & \dots & \frac{\partial f_2}{\partial x_p}(a) \\ \vdots & & \vdots \\ \frac{\partial f_q}{\partial x_1}(a) & \dots & \frac{\partial f_q}{\partial x_p}(a) \end{pmatrix}$$

$$\text{det } J_f(a) = \det \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_p} \\ \vdots & & \vdots \\ \frac{\partial f_q}{\partial x_1} & \dots & \frac{\partial f_q}{\partial x_p} \end{pmatrix} = \det J_f(a)$$

$d(x_1, \dots, x_p)$

- P**
- 1) f differentiable in $a \Rightarrow f$ cont. and
 - 2) f differentiable in $a \Rightarrow \exists \frac{\partial f}{\partial x_i}(a)$

Th (cond. suficiente, dar nu merita)
 $D \subset \mathbb{R}^n$ $\exists V \in U(a)$ at $\exists$ derivata partiala
 $f \in V$, continue in $a \Rightarrow$ dif. in a

Ex $f: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $f(x, y, z) = (xy + yz + xz, xzy)$

Sa se scrie matricea Jacobiană
 in $(1, 1, 2)$.

Sol:
 $J_f(x, y, z) = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \end{pmatrix}$

$= \begin{pmatrix} y+z & x+z & x+y \\ yz & xz & xy \end{pmatrix}$

$J_f(1, 1, 2) = \begin{pmatrix} 3 & 3 & 2 \\ 2 & 2 & 1 \end{pmatrix}$

① Ex $f: \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $f(x, y) = (x^2 + y^2, xy, \frac{x}{y^2 + 1})$. Calc m. j. in $(0, 0)$.

2. $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = \sqrt{x^2 + 2y^2 + 1}$.

a) $Df(1, 2) = ?$

b) $Df(1, 2)(-1, 3) = ?$

Sol:

a) $Df(1, 2) = \frac{\partial f}{\partial x}(1, 2) dx + \frac{\partial f}{\partial y}(1, 2) dy$

$\frac{\partial f}{\partial x}(x, y) = \frac{1}{2\sqrt{x^2 + 2y^2 + 1}}$. $\frac{\partial f}{\partial y} = \frac{2y}{\sqrt{x^2 + 2y^2 + 1}}$

$\Rightarrow \frac{\partial f}{\partial x}(1, 2) = \frac{1}{\sqrt{10}}$

$\frac{\partial f}{\partial y}(x, y) = \frac{2y}{\sqrt{x^2 + 2y^2 + 1}} \Rightarrow$

$\frac{\partial f}{\partial y}(1, 2) = \frac{4}{\sqrt{10}}$

$$\Rightarrow D\varphi(1,2) = \frac{1}{\sqrt{10}} \frac{\partial \varphi}{\partial x} + \frac{2}{\sqrt{10}} \frac{\partial \varphi}{\partial y} = 0$$

$$\begin{aligned} \text{b) } D\varphi(1,2)(-1,3) &= \\ &= \frac{1}{\sqrt{10}} \cdot (-1) + \frac{2}{\sqrt{10}} \cdot 3 = \frac{1}{\sqrt{10}} \end{aligned}$$

$$\textcircled{1} \quad \varphi: \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$\varphi(x,y,z) = \ln(x^2 y^2 + z^2 + 2)$$

$$\text{a) } D\varphi(1,2,-1) = ?$$

$$\text{b) } D\varphi(1,2,-1)(0,0,1) = ?$$

$\mathbb{R}$	$\mathbb{R}^2$
$\varphi(x) = \begin{cases} \dots, & x \neq 0 \\ \dots, & x = 0 \end{cases}$	$\varphi(x,y) = \begin{cases} \dots, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$
<u>Cont.</u> : $\lim_{x \rightarrow 0} \varphi(x) = \varphi(0)$	<u>Cont.</u> : ? $\Delta \checkmark$
<u>Deriv.</u> : $\varphi'(x) = \begin{cases} \dots, & x \neq 0 \\ \text{def}, & x = 0 \end{cases}$	∇ derivative ∇ partial

$$\varphi(x_n, y_n) \xrightarrow{z} (0,0) \Rightarrow$$

$$\Rightarrow \varphi(x_n, y_n) \xrightarrow{z} \varphi(0,0) = 0$$

$$0 \leq |\varphi(x_n, y_n)| \leq \epsilon$$

$$0 \leq |f(x,y)| \leq \dots \quad \text{?}$$

Consider

$$(x_n, y_n) \rightarrow (0,0)$$

$$\left(\frac{1}{n}, \frac{1}{n}\right)$$

$$\left(\frac{1}{n}, \frac{1}{\sqrt{n}}\right) \text{ at } f(x_n, y_n) \not\rightarrow 0 = f(0,0)$$

$\exists$ deriv. positive

$$\frac{\partial f}{\partial x}(x,y) = \begin{cases} (\dots)'x, & (x,y) \neq (0,0) \\ \text{def} & (x,y) = (0,0) \end{cases}$$

differentiability (in $(0,0)$)

$$(dc \exists Df(0,0) = \frac{\partial f}{\partial x}(0,0)dx + \frac{\partial f}{\partial y}(0,0)dy$$

$$? \lim_{\|(x,y)-(0,0)\| \rightarrow 0} \frac{f(x,y) - f(0,0) - Df(0,0)(x,y)}{\|(x,y)-(0,0)\|} = 0$$

1. Studiate cont. fct.

$$1) f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x,y) = \begin{cases} \frac{\partial^2 x^2 y}{\sqrt{x^4+y^4}}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

$$2) f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x,y) = \begin{cases} (1+2xy) \frac{x^2+y^2}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

$$3) f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x,y) = \begin{cases} \frac{\sin(x^2+y^2)}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

$$4) f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x,y) = \begin{cases} \frac{\ln(1+xy)}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

f cont. $\lim_{n \rightarrow \infty} (a, b_n) \Rightarrow f(x_n, y_n)$

$$\lim_{n \rightarrow \infty} (x_n, y_n) \rightarrow (a,b) \Rightarrow$$

$$\Rightarrow f(x_n, y_n) \rightarrow f(a,b)$$

SOL:

1) Find if cont. f on $\mathbb{R}^2 \setminus \{(0,0)\}$
Cont. in $(0,0)$:

Let $(x_n, y_n) \rightarrow (0,0)$

$$0 \leq |f(x_n, y_n)| \leq \left| \frac{3x_n^2 y_n}{\sqrt{x_n^4 + y_n^4}} \right|$$

$$\leq \left| \frac{3x_n^2 y_n}{\sqrt{x_n^4}} \right| = 3 \cdot |y_n|$$

$\rightarrow 0$

$f(0,0) \Rightarrow f$ cont. in $(0,0) \Rightarrow$

$\Rightarrow f$ cont. on $\mathbb{R}^2$

2) Find if cont. f on $\mathbb{R}^2 \setminus \{(0,0)\}$
in $(0,0)$:

Let $(x_n, y_n) \rightarrow (0,0)$

$$0 \leq |f(x_n, y_n)| \leq \left| (1 + 2x_n y_n^2) \frac{2x_n y_n^2}{x_n^3 + y_n^3} \right|$$

$$\left((1 + 2x_n y_n^2) \frac{2x_n y_n^2}{x_n^3 + y_n^3} \right) \neq 0$$

$$\varphi = \lim_{n \rightarrow \infty} \frac{2x_n y_n^2}{x_n^3 + y_n^3} = ?$$

Let $(x_n, y_n) = \left(\frac{1}{n}, \frac{1}{n} \right) \rightarrow (0,0)$

$\Rightarrow \varphi = 1$

$(x_n', y_n') = \left(\frac{1}{n}, \frac{1}{n} \right) \rightarrow (0,0)$

$\Rightarrow \varphi = \frac{1}{2} \Rightarrow \varphi \neq$

$\Rightarrow f$ not cont. in $(0,0)$.

$$3) f(x, y) = \begin{cases} \sin(x^3 + y^3), & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$\text{Die } (x_n, y_n) \rightarrow (0, 0)$$

$$\begin{aligned} \rho = |f(x_n, y_n)| &= \left| \sin(x_n^3 + y_n^3) \right| \\ &= \left| \frac{\sin(x_n^3 + y_n^3)}{x_n^3 + y_n^3} \right| \cdot \left| \frac{x_n^3 + y_n^3}{x_n^3 + y_n^3} \right| \\ &\leq \left| \frac{\sin \dots}{\dots} \right| \cdot \left(\left| \frac{x_n^3}{x_n^3 + y_n^3} \right| + \left| \frac{y_n^3}{x_n^3 + y_n^3} \right| \right) \\ &\quad \leq \underbrace{\left| \frac{x_n^3}{x_n^3 + y_n^3} \right|}_{\rightarrow 0} + \underbrace{\left| \frac{y_n^3}{x_n^3 + y_n^3} \right|}_{\rightarrow 0} \\ &\quad \leq \underbrace{\left| \frac{x_n^3}{x_n^3 + y_n^3} \right|}_{\rightarrow 0} + \underbrace{\left| \frac{y_n^3}{x_n^3 + y_n^3} \right|}_{\rightarrow 0} \end{aligned}$$

$$\Rightarrow f(x_n, y_n) \rightarrow 0 \Rightarrow$$

$$\Rightarrow f \text{ cont. in } \mathbb{R}^2$$

$$4) f(x, y) = \begin{cases} \frac{1}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$\text{Die } (x_n, y_n) = \left(\frac{1}{n}, \frac{1}{n} \right) \rightarrow (0, 0)$$

$$\Rightarrow f\left(\frac{1}{n}, \frac{1}{n}\right) = \frac{1}{\sqrt{\frac{1}{n^2} + \frac{1}{n^2}}} = \frac{1}{\sqrt{\frac{2}{n^2}}} = \frac{n}{\sqrt{2}} \rightarrow \infty$$

$$(x_n, y_n) = \left(\frac{1}{n}, \frac{1}{n} \right) \rightarrow (0, 0)$$

$$\Rightarrow f\left(\frac{1}{n}, \frac{1}{n}\right) = \frac{1}{\sqrt{\frac{1}{n^2} + \frac{1}{n^2}}} = \frac{1}{\sqrt{\frac{2}{n^2}}} = \frac{n}{\sqrt{2}} \rightarrow \infty$$

$$\Rightarrow f \text{ not cont. in } (0, 0)$$

2. Fie $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $(x, y) \neq (0, 0)$

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

a) Cont. f ?

b) $\exists$ deriv. partiiale?

c) f are ∇f dif. în originea

d) f dif. pe $\mathbb{R}^2 \setminus \{(0, 0)\}$

e) f are deriv. după orice vector $v \neq 0_{\mathbb{R}^2} = (0, 0)$?

Sol:

a) f cont. pe $\mathbb{R}^2 \setminus \{(0, 0)\}$.

Cont. $\text{în } (0, 0)$:

$$\text{Fie } (x_n, y_n) \rightarrow (0, 0).$$

$$\text{Vrem } f(x_n, y_n) \rightarrow f(0, 0) = 0$$

$$0 \leq |f(x_n, y_n)| = \left| \frac{x_n y_n}{\sqrt{x_n^2 + y_n^2}} \right| \leq \left| \frac{x_n y_n}{\sqrt{x_n^2}} \right| = |y_n| \rightarrow 0$$

$$\Rightarrow f(x_n, y_n) \rightarrow 0.$$

$$\Rightarrow f \text{ cont. } \text{în } (0, 0) \Rightarrow$$

$$\Rightarrow f \text{ cont. pe } \mathbb{R}^2.$$

$$\begin{aligned} \text{b) } \frac{\partial f}{\partial x}(0, 0) &= \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \\ &= \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0. \end{aligned}$$

$$\text{Pentru } (x, y) \neq (0, 0) \Rightarrow$$

$$\begin{aligned} \Rightarrow \frac{\partial f}{\partial x}(x, y) &= \frac{y\sqrt{x^2+y^2} - xy \cdot \frac{1}{2\sqrt{x^2+y^2}} \cdot 2x}{x^2+y^2} \\ &= \frac{y(x^2+y^2) - x^2 y}{(x^2+y^2)\sqrt{x^2+y^2}} = \frac{y^3}{\sqrt{x^2+y^2}} \end{aligned}$$

$$\Rightarrow \frac{\partial f}{\partial x}(x,y) = \begin{cases} \frac{xy^3}{\sqrt{x^2+y^2}} \cdot (x^2+y^2) & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

$$\text{simetria } \frac{\partial f}{\partial y}(x,y) = \begin{cases} \frac{x^3y}{\sqrt{x^2+y^2}} \cdot (x^2+y^2) & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

$\Rightarrow \nabla f$ em todo ponto de $\mathbb{R}^2$.

$\Rightarrow f$ no  dif. em origem

Pp. f dif. em $(0,0) \Leftrightarrow$

$$\begin{aligned} \Rightarrow Df(0,0) &= \frac{\partial f}{\partial x}(0,0)dx + \frac{\partial f}{\partial y}(0,0)dy \\ &= 0dx + 0dy = 0 \\ \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - Df(0,0)}{\|(x,y) - (0,0)\|} &= 0 \end{aligned}$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2} = 0$$

$$\text{Se } (x_m, y_m) = \left(\frac{1}{m}, \frac{1}{m}\right) \rightarrow (0,0) \Rightarrow$$

$$\Rightarrow \lim_{m \rightarrow \infty} \frac{x_m y_m}{x_m^2 + y_m^2} = \lim_{m \rightarrow \infty} \frac{\frac{1}{m^2}}{\frac{2}{m^2}} = \frac{1}{2} \neq 0$$

$\Rightarrow f$ no  diferencivel em $(0,0)$

b) f dif. p $\mathbb{R}^2 \setminus \{(0,0)\}$

$\mathbb{R}^2 \setminus \{(0,0)\}$ m.d.

$\exists$ derivadas parciais cont. em todo

ponto $(x,y) \in \mathbb{R}^2 \setminus \{(0,0)\}$

$\Rightarrow f$ diferencivel p $\mathbb{R}^2 \setminus \{(0,0)\}$

c) Se $v = (v_1, v_2) \neq (0,0)$, $a = (0,0)$

$$\lim_{t \rightarrow 0} \frac{f((0,0) + t(v_1, v_2)) - f(0,0)}{t} =$$

$$\begin{aligned}
 &= \lim_{t \rightarrow 0} \frac{f(t v_1, t v_2) - 0}{t} = \\
 &= \lim_{t \rightarrow 0} \frac{t v_1 \cdot t v_2}{\sqrt{t^2 v_1^2 + t^2 v_2^2}} = \\
 &= \lim_{t \rightarrow 0} \frac{t \cdot v_1 v_2}{t \sqrt{v_1^2 + v_2^2}} \neq
 \end{aligned}$$

$\Rightarrow$ f nu are derivat part, dupa orice vector.

① 1. $f: \mathbb{R}^2 \rightarrow \mathbb{R}$

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

- a) — 1 —
- b) — 1 —
- c) — 1 —
- d) — 1 —
- e) $\checkmark$

2. Fie $f: \mathbb{R}^2 \rightarrow \mathbb{R}$

$$f(x, y) = \begin{cases} (x^2 + y^2) \cdot \sin \frac{1}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

- a) cont. f
- b) f este dif. in $(0, 0)$, deci f dif. pe $\mathbb{R}^2$

c) $\exists \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ in orice pt, dar nu sunt continue in $(0, 0)$.